

Three Physical Walls Create Four Deployment Paradigms

Light Barrier



Speed of Light in Fiber

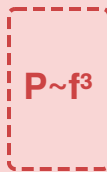
~200,000 km/s

CA to VA round-trip
~36 ms minimum

Sub-10 ms systems cannot
use distant cloud servers

Creates Edge + TinyML

Power Wall



Thermodynamic Limit

$$P = C \times V^2 \times f, V \sim f$$

2x frequency = 8x power
Mobile TDP: 3-5 W

Datacenter models cannot
run on battery devices

Creates Mobile ML

Memory Wall



Compute-Memory Gap

1.6x/yr vs 1.2x/yr = 1.33x/yr

Data movement dominates
TinyML: 256 KB SRAM

Moving data often costlier
than computing on it

Creates TinyML limits

These walls are physics, not engineering --- they cannot be optimized away

Physical laws create the four deployment paradigms: Cloud, Edge, Mobile, TinyML